Getting Started

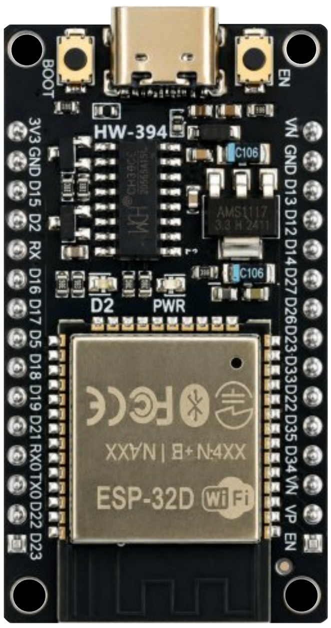
The Getting Started with IoT course is divided into two sections.

Part 1
Covers the assembly and testing of the ESP32 IoT Controller.

Part 2
Focuses on interfacing and programming sensors and actuators.

This document introduces the components included in the course and briefly explains their functions. At this stage, learners only need to identify each component and understand its basic purpose. Each sensor and actuator will be studied in greater detail during the practical programming exercises.

Part 1 - ESP32 IoT Controller

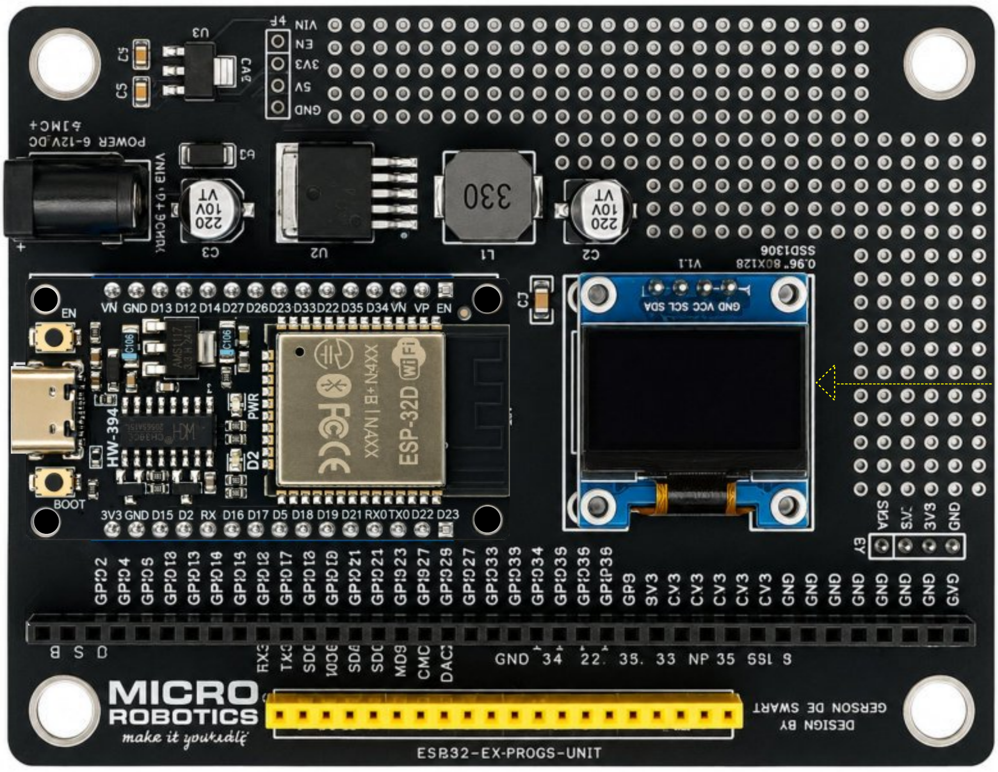


The ESP32 30-pin development module is the most popular, IoT controller. It features integrated Wi-Fi, Bluetooth, digital and analogue inputs, PWM outputs, and multiple communication interfaces.

The module can connect sensors,
displays, and actuators while processing data and communicating with online services.

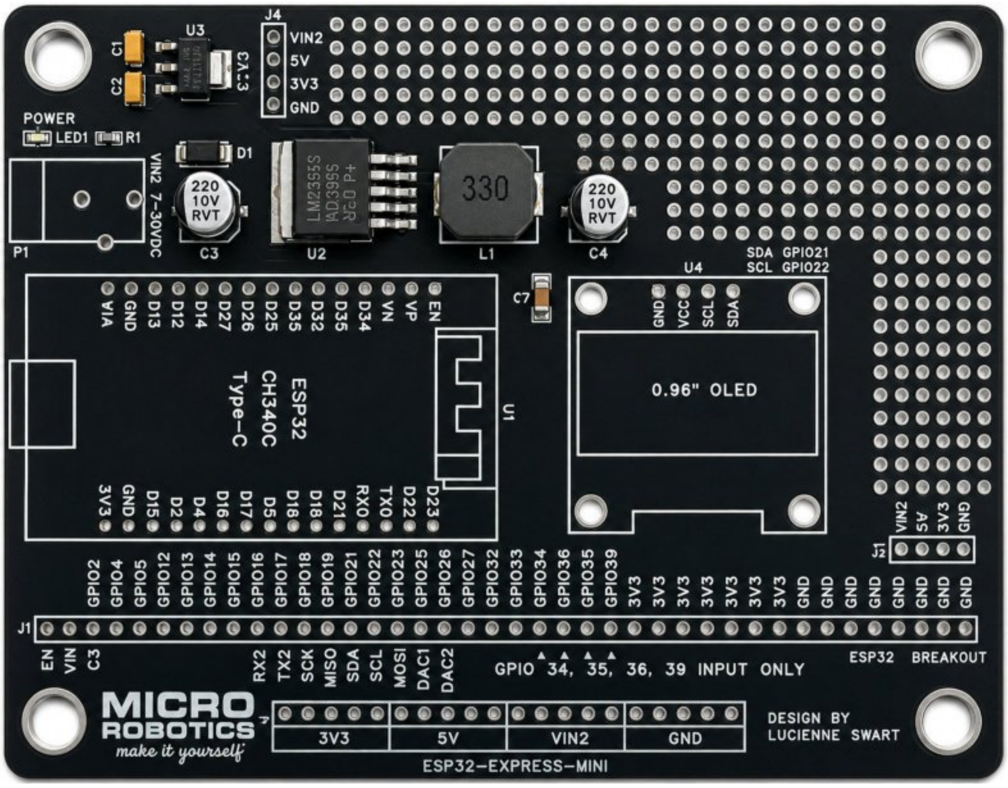
Part 1 - ESP32 Development Board

To help newcomers explore electronics, we developed a complete IoT board with access to all ESP32 pins, an OLED display, and integrated power regulation for reliable learning and prototyping.



Fully Assembled ESP32 IOT Controller

Part 1 - IOT Controller Components

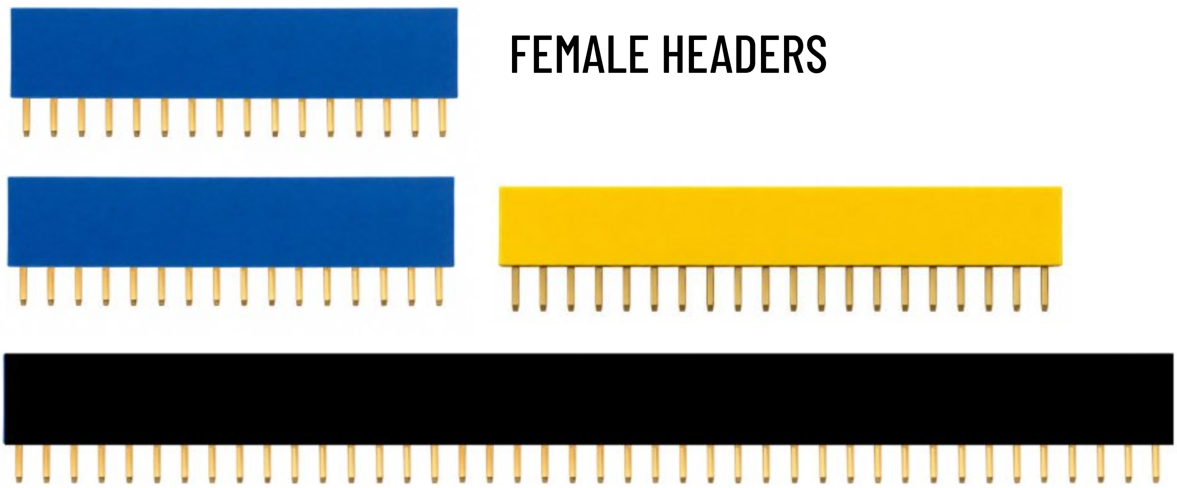


IOT Controller PC Board

ESP32 Controller

DC JACK

OLED Display

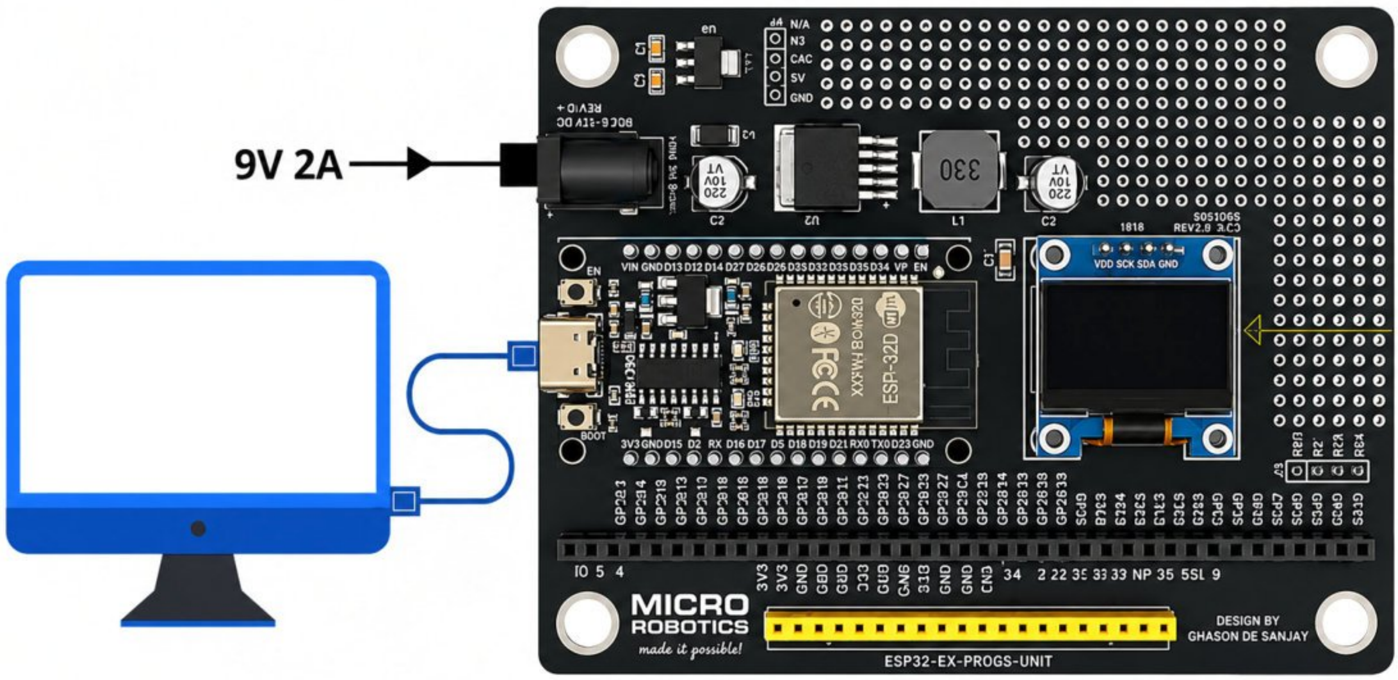


FEMALE HEADERS

Part 1 - Final Testing and Software Setup

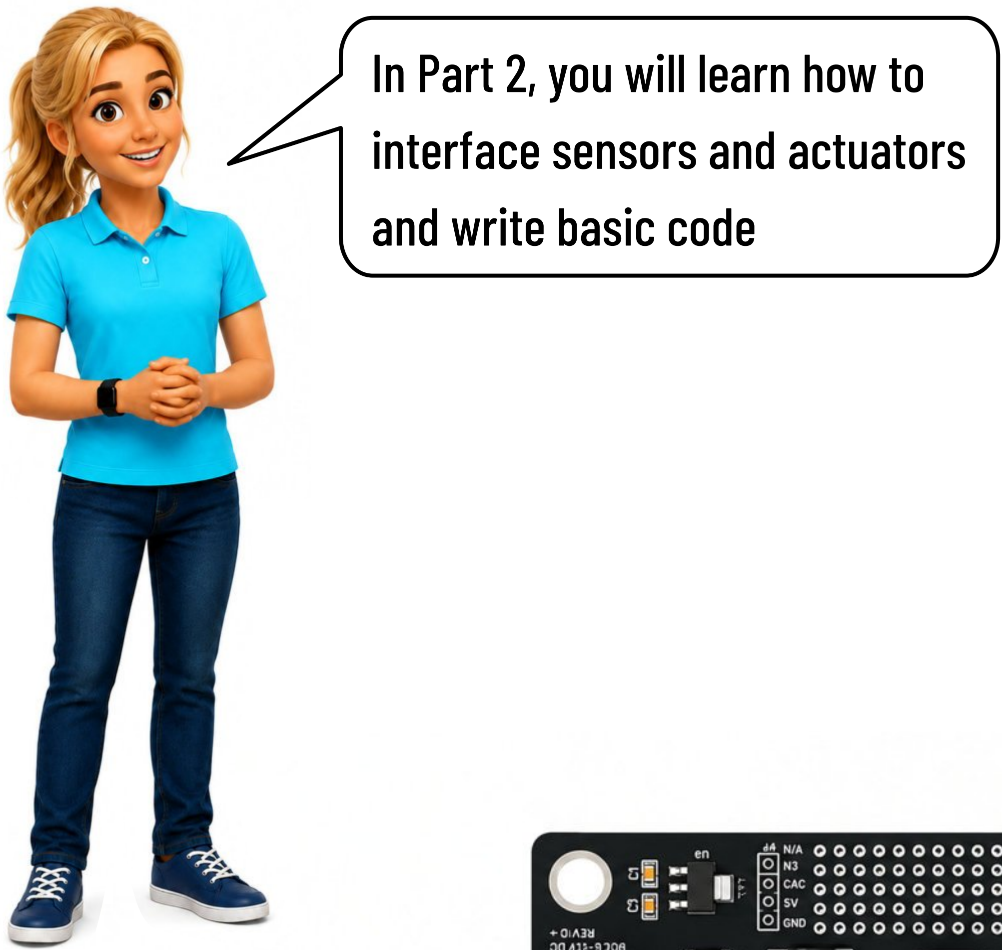
With the ESP32 IoT Controller assembled, we can now test the board and OLED display.

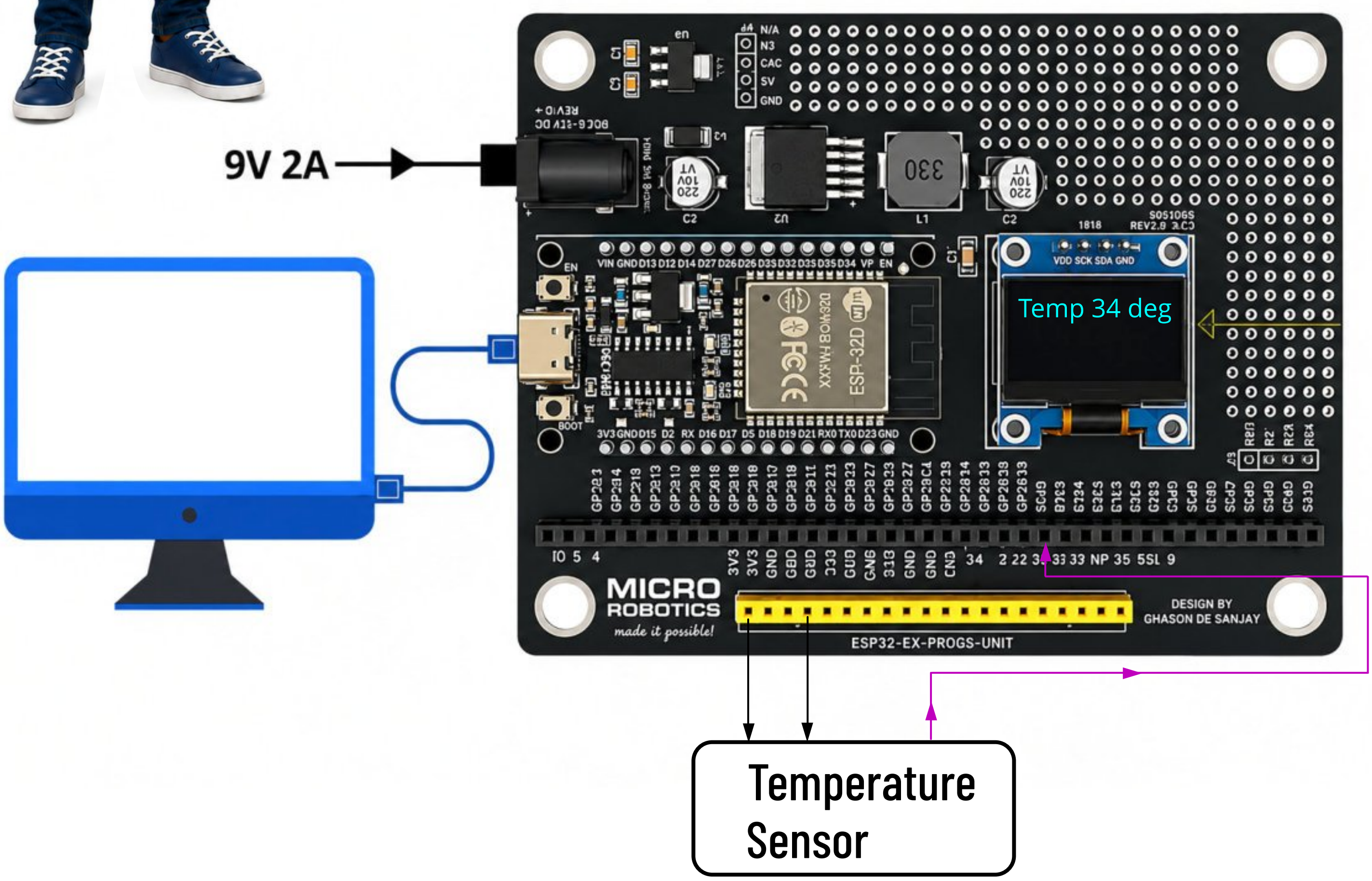
- Downloading latest software IDE
- Testing the onbaord LED
- Downloading OLED driver
- Display your first “Hello World” message



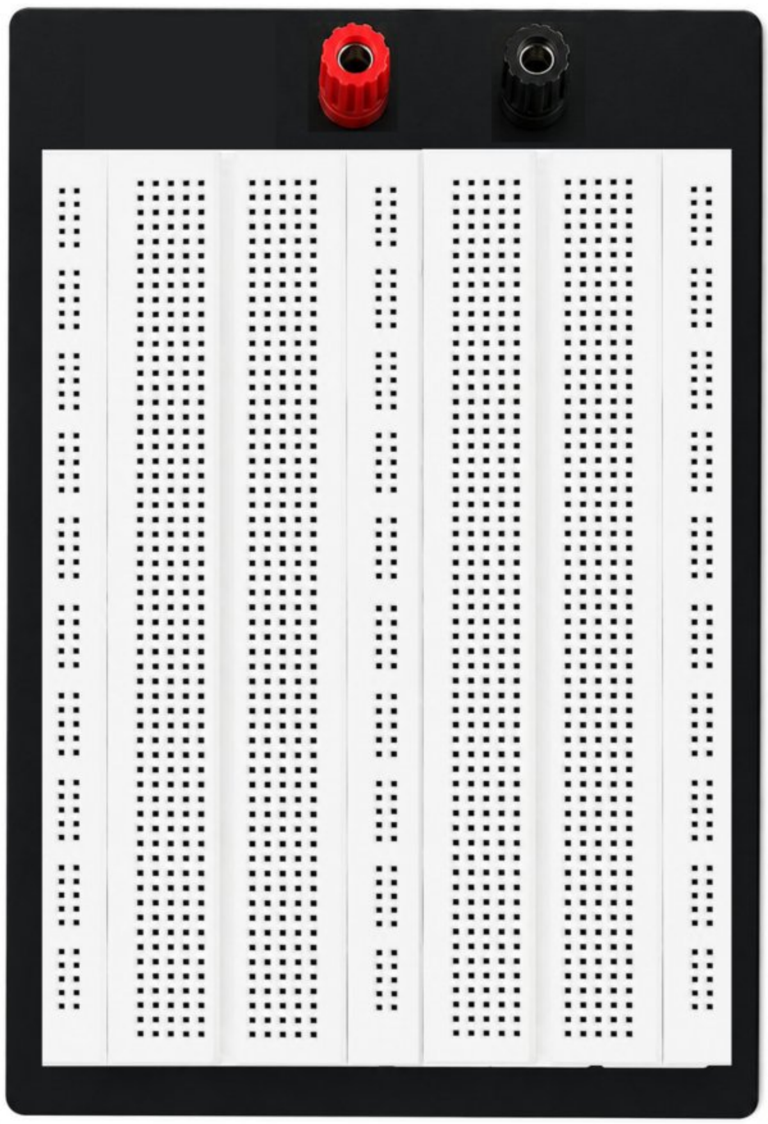
Getting Started - Part 2

Interfacing Sensors and Actuators

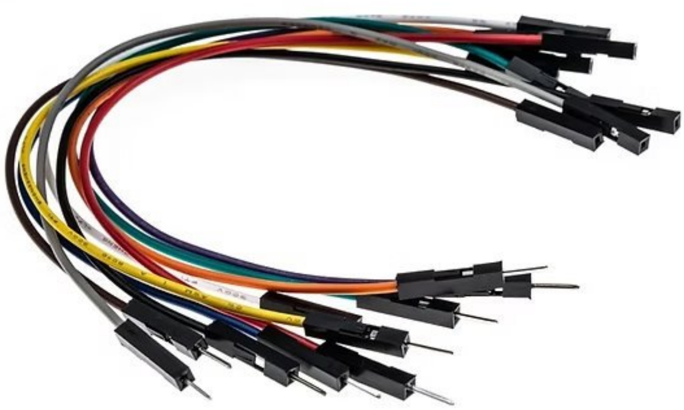




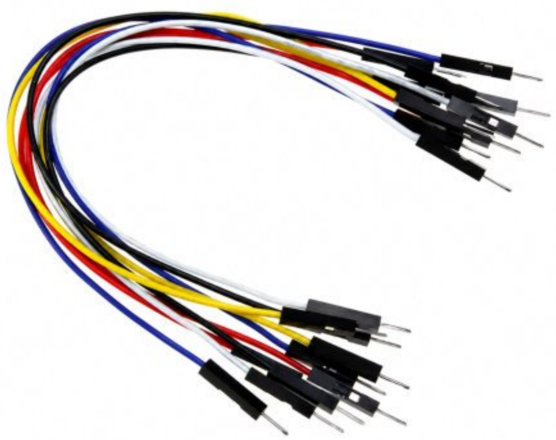
1. Breadboard and Jumpers



Male Male Jumpers



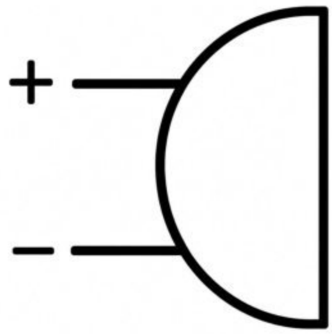
Male Female Jumpers



Male Male Jumpers

2. Passive Buzzer 3.3V

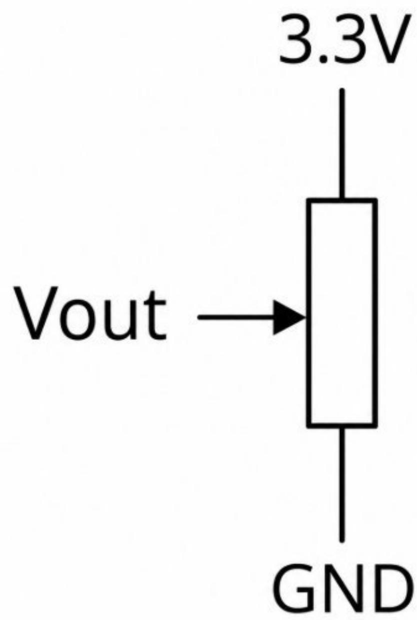
These 3.3 V buzzer converts electrical signals into sound, they produce a fixed tone when powered, They are commonly used with ESP32 projects for alarms, status notifications, timers, button feedback, fault warnings and simple sound effects.



SYMBOL

3. Potentiometer (Varaible Resistor)

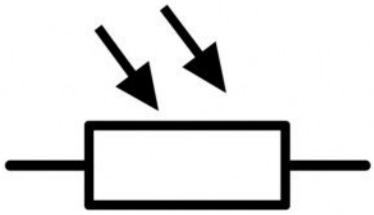
A potentiometer is a variable resistor used to adjust voltage or electrical signals. Since many sensors provide analogue outputs, potentiometers are useful for simulating sensor signals when testing and debugging code.



SYMBOL

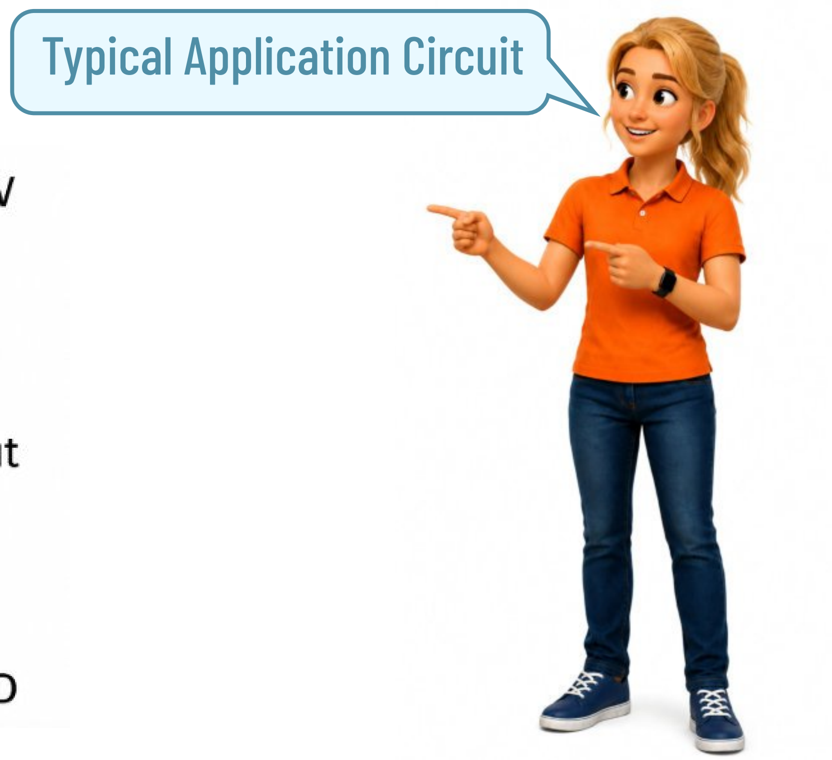
5. LDR - Light Dependant Resistor 10K

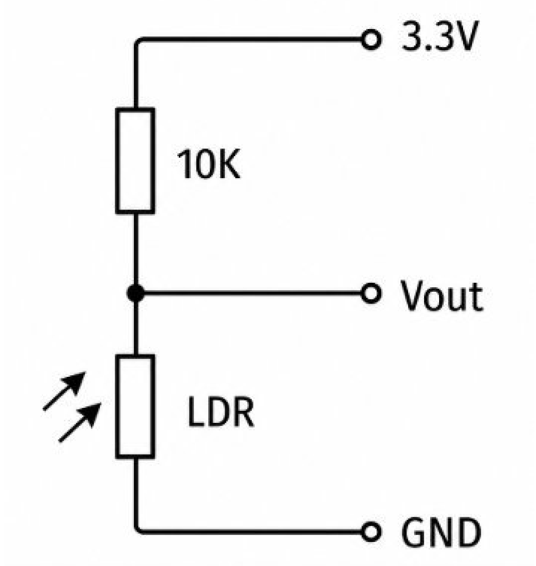
An LDR (Light Dependent Resistor) changes resistance according to light level. The ESP32 reads this change through an analogue input. LDRs are commonly used in automatic lights, brightness sensors, alarms and projects that detect day and night.



SYMBOL

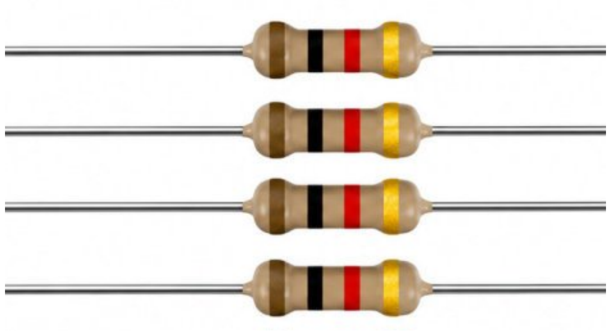
In darkness, resistance rises above 10 kΩ. In bright sunlight, it drops below 1 kΩ, depending on the LDR.





6. Resistors 1K and 10K Value

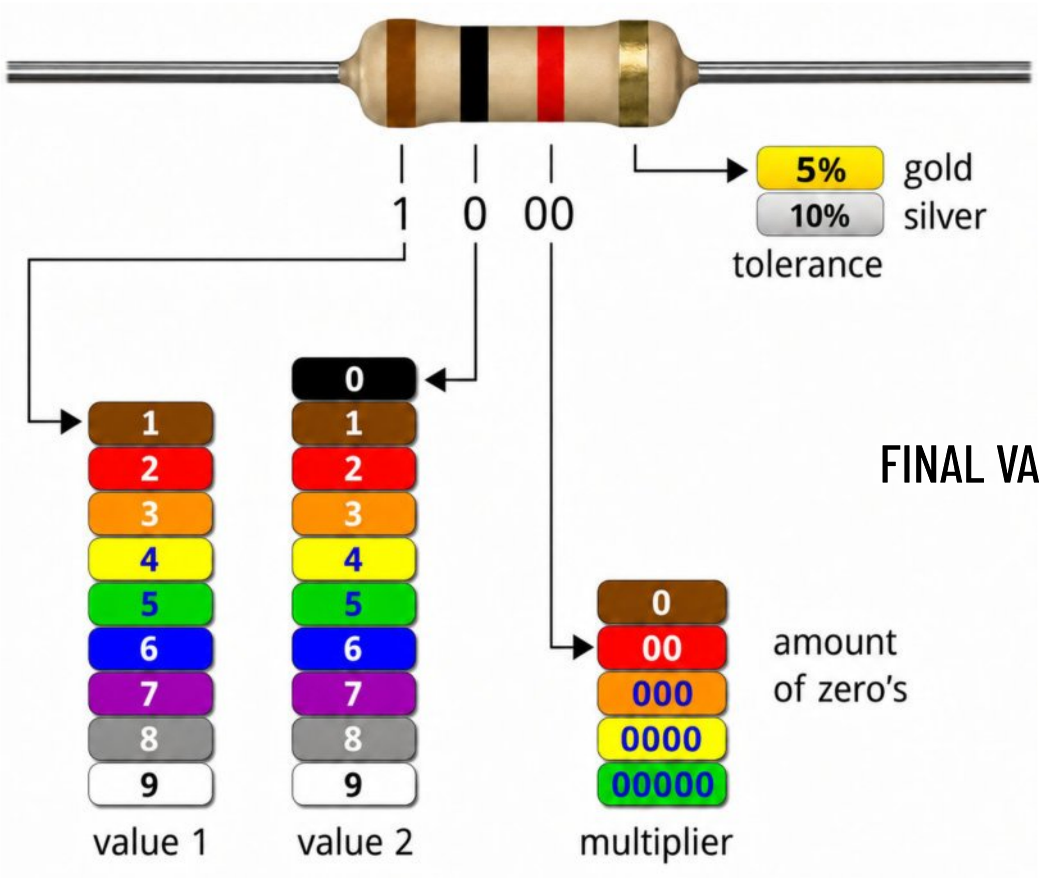
A resistor limits electrical current and controls voltage to protect components in an electronic circuit.



1K OHM

10K OHM

In this section, you will learn how to identify resistor values and how to apply Ohm's Law.

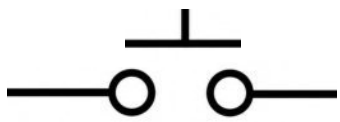


FINAL VALUE = 10K OHM

7. 18B20 Temperature Sensor

The DS18B20 is a digital temperature sensor that communicates via a 1-Wire interface. It provides accurate measurements and is widely used in industrial control, refrigeration, HVAC and environmental monitoring.

8. Switches



9. Moisture Soil Sensor

It’s a soil moisture sensor, specifically a probe-type resistive soil moisture sensor. The two metal rods go into the soil and measure conductivity, which changes with water content. It’s used for plant watering, irrigation control, and soil monitoring.

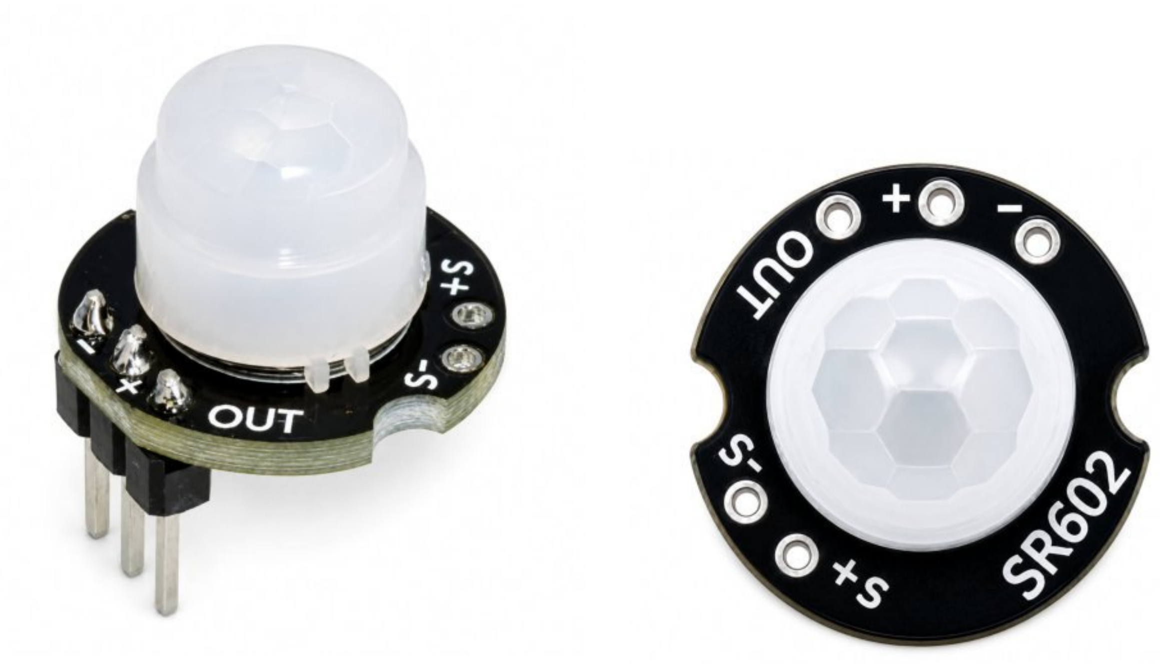
10. RC Servo Motor (Actuator)

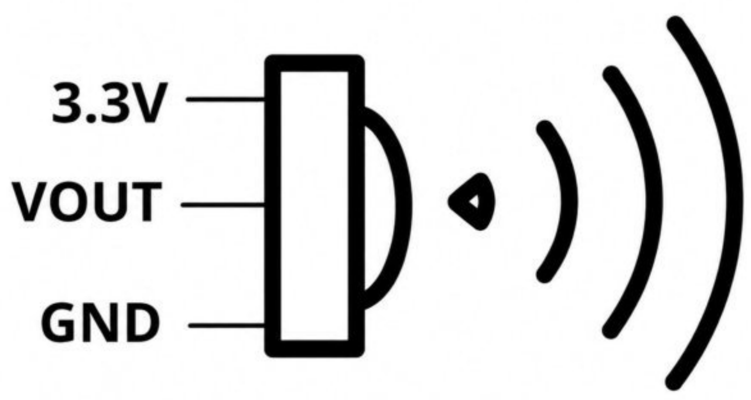
An RC servo motor precisely controls angular position using PWM signals. It combines a motor, gearbox, position sensor, and control circuit.

The FS90 servo supports up to approximately 180° rotation

11. SR602 PIR Sensor (Passive Infrared Sensor)

The SR602 PIR sensor detects movement by sensing changes in infrared heat from people or animals. It operates from about 3.3–15 V, provides a digital output, and is commonly used in alarms and ESP32 projects.

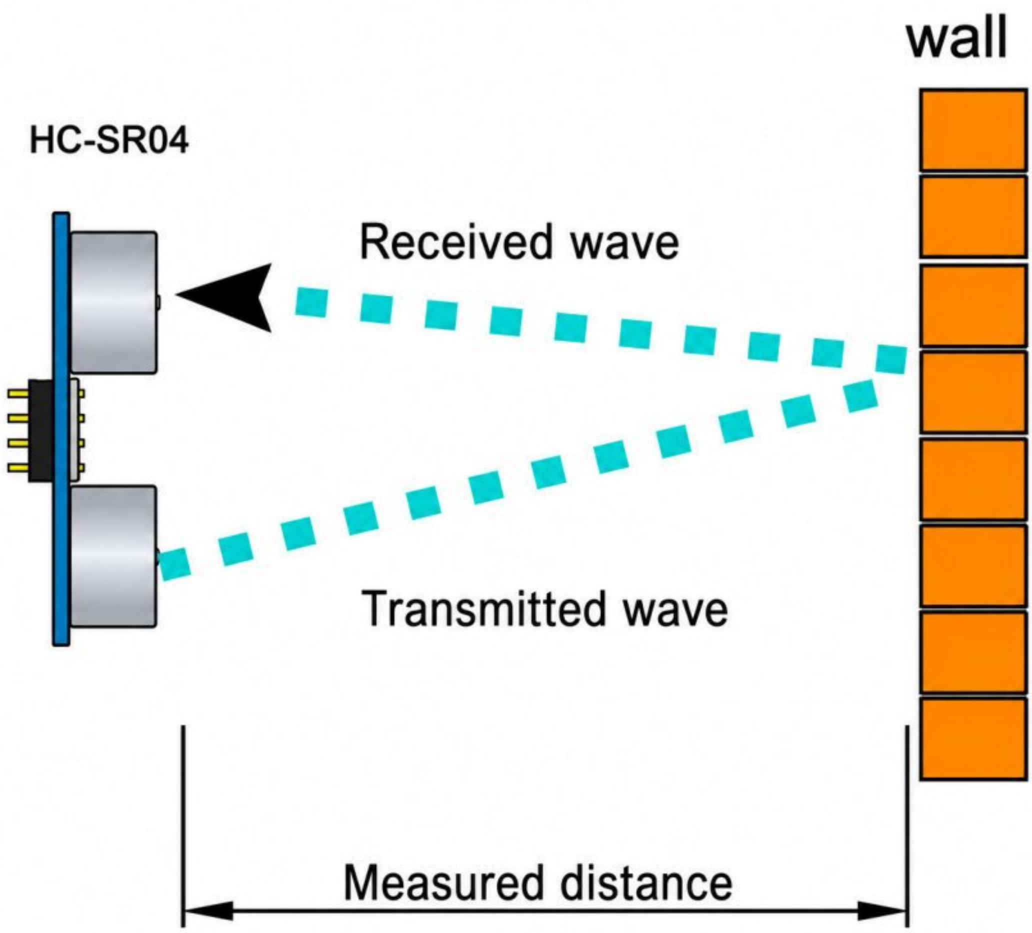




12. HC-SR04 (Distance Sensor)

The HC-SR04 is an ultrasonic distance sensor. It measures distance by sending a high-frequency sound pulse and timing the echo return. It is widely used with Arduino and ESP32 for obstacle detection, level sensing, and simple distance measurement.

VCC	5V
Trig	Trigger Input Pin
Echo	Echo Output Pin
GND	GND

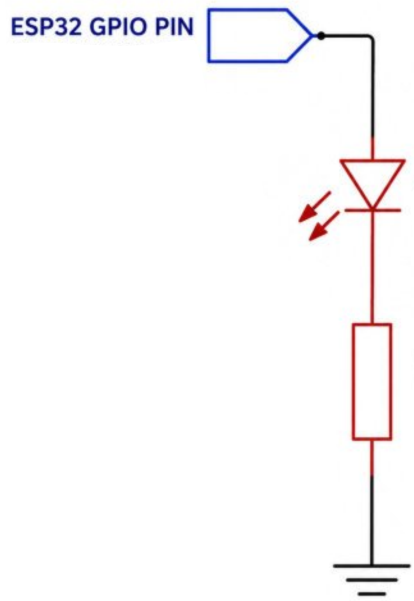


13. Relay Module

Relay modules allow an ESP32 to safely switch higher-voltage or higher-current loads. They provide electrical isolation and signal amplification allow to switch external loads up to 250V 10A

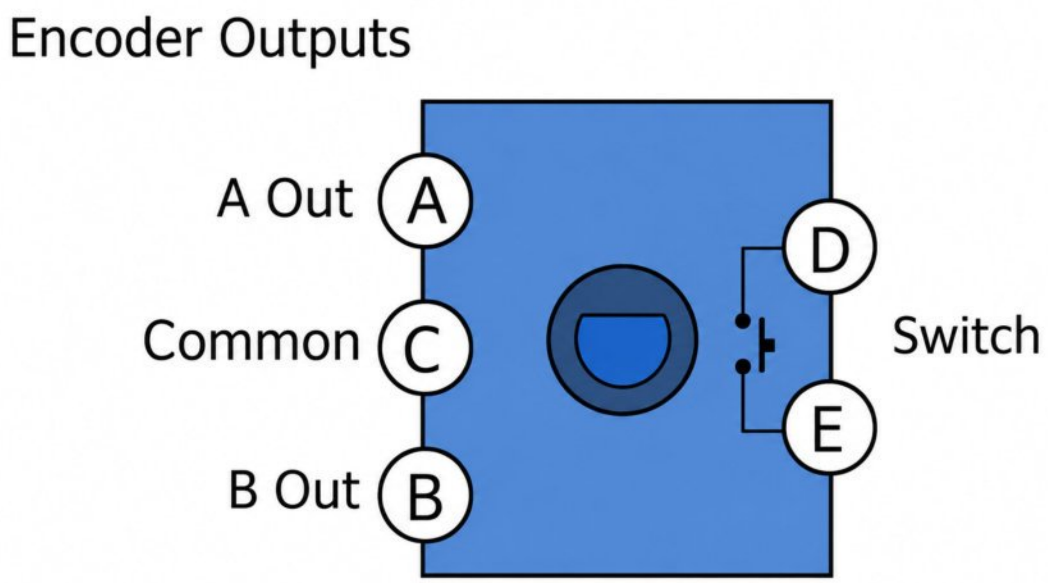
14. LED - Light Emitting Diodes

LEDs provide visual status and warning indications. In ESP32 projects, we will calculate the correct current-limiting resistor to protect the GPIO pin from excessive current.



15. Rotory Encoder

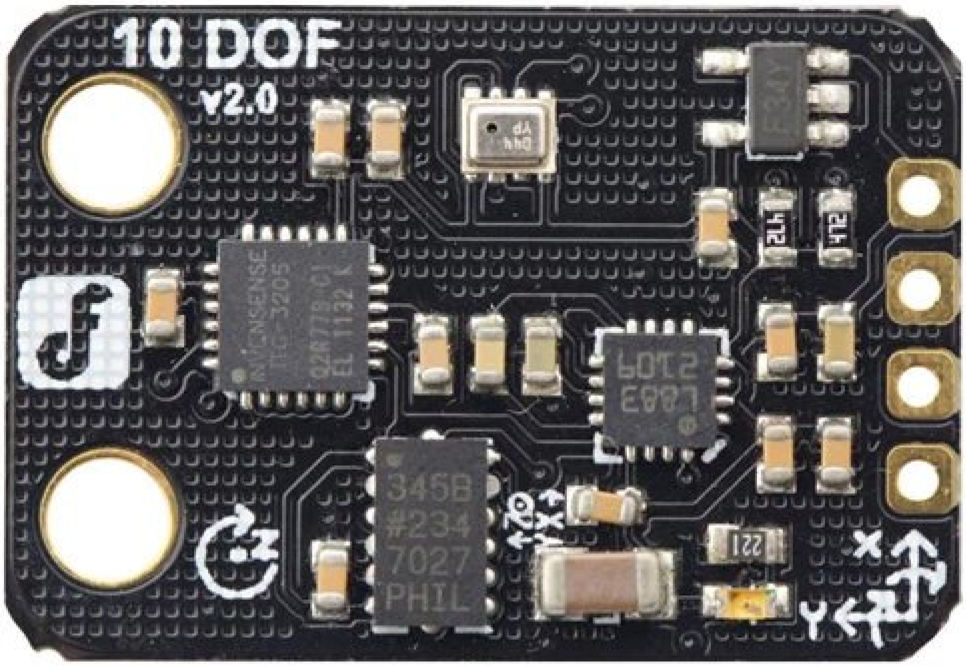
A rotary encoder converts shaft rotation into digital pulses. In ESP32 projects, it can control menus, adjust values, select modes, measure position, or detect rotation direction and speed.

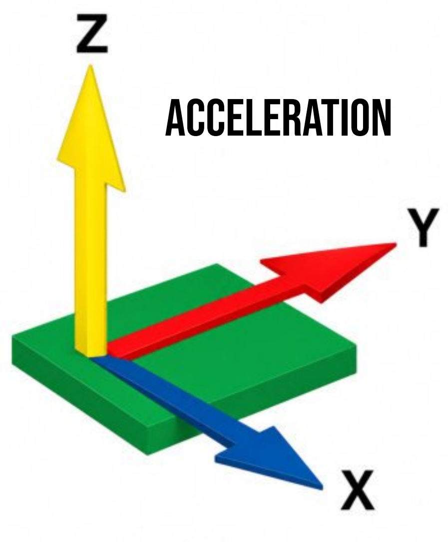


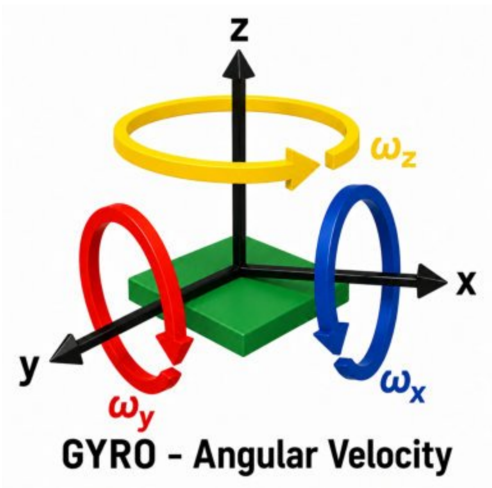
16. Sensor 10-DoF sensor (Acceleration, Gyro, Magnetometer & Barometric)

A 10-DoF sensor measures motion, orientation, altitude and heading using a 3-axis accelerometer, 3-axis gyroscope, 3-axis magnetometer and barometric pressure sensor.

Manufacturer Website : https://www.dfrobot.com/product-818.html
SKU: SEN0140



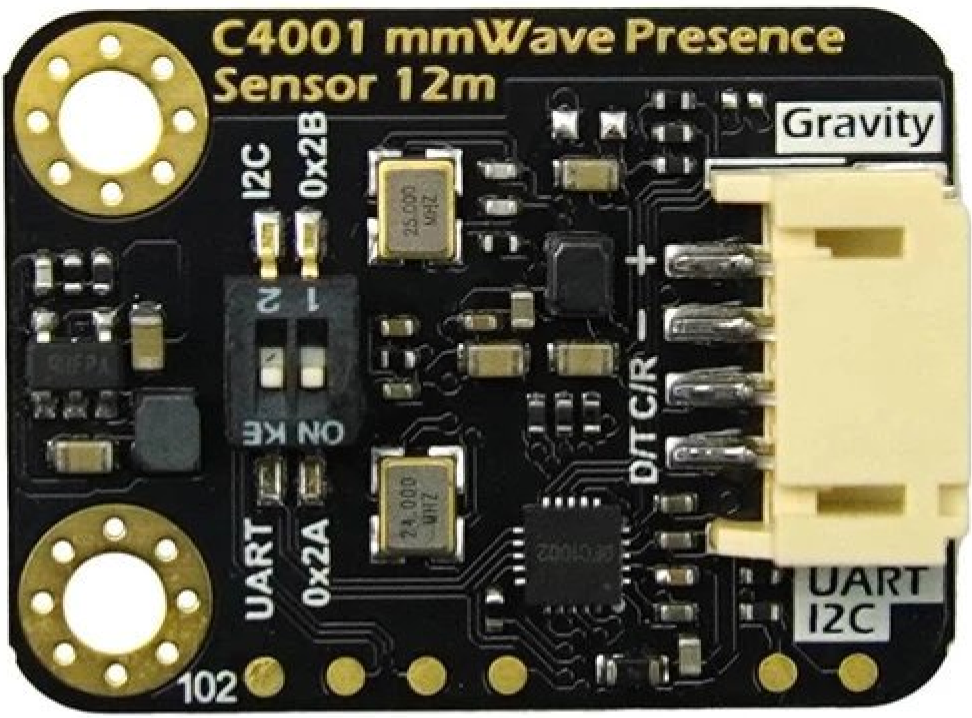




17. Radar Sensor - Distance/Speed Measurement - 12 24 Meters

The DFRobot SEN0610 is a 24 GHz mmWave radar sensor that detects stationary people up to 8 m and movement up to 12 m. It can measure distance and speed and is useful for smart lighting, security, occupancy detection and ESP32 automation projects

Manufacturer Website : https://www.dfrobot.com/product-2795.html
SKU: SEN0610



18. DHT22 Temperature and Humidity Sensor

The DHT22 is a digital sensor that measures air temperature and relative humidity. It sends calibrated readings to an ESP32 using one data pin.
It is commonly used in weather stations, greenhouses, HVAC systems and environmental monitoring projects.